Reg. No. | | | | | | | | | | |

[14EC36]

THIRD SEMESTER B.TECH EXAMINATION DECEMBER - 2015

ELECTRONIC MEASUREMENTS AND INSTRUMENTATION

Time:3 Hours **Maximum Marks:70**

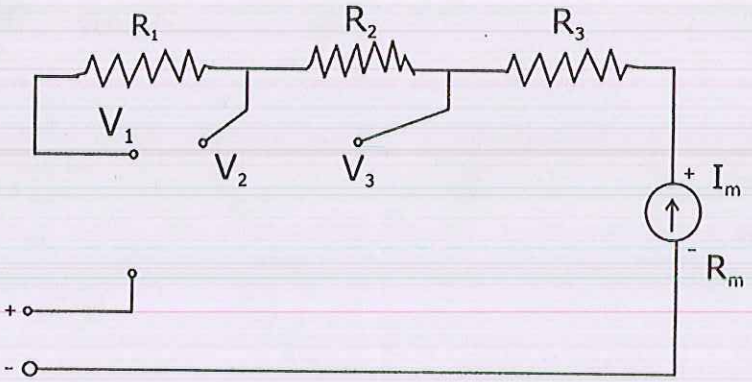
Instructions: Answer any **FIVE** full questions. Selecting atleast TWO from each part.

PART - A

1. a. Define the following terms: **03**
   - i. Instrument
   - ii. Accuracy
   - iii. Resolution

   b. Explain how do we convert a basic D'Arsonval movement to form a dc Voltmeter. **05**

   c. Convert a basic D'Arsonval movement with an Internal resistance of 100Ω and a full scale Deflection of 10mA into a multi range dc voltmeter with ranges from 0-5V, 0-50V, 0-100V. **06**



2. a. With the help of neat block diagram, explain a standard signal generator. **07**

   b. Explain function generator with the help of a neat block diagram. **07**

3. a. With the help of a neat circuit, explain the operation of a Basic wave analyzer. **07**

   b. Explain with a neat block diagram the working of a Heterodyne wave analyzer. **07**

4. a. List and explain the important CRT features. **07**

   b. With the help of neat block diagram explain Dual Beam CRO. **07**

PART-B

5. a. With the help of neat block diagram explain the operation of sampling Oscilloscope. **07**

   b. Explain Alternate mode and chop modes of operations with respect to dual trace Oscilloscope. **07**

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6.	**a.**	With the help of neat circuit, explain Maxwell's bridge for the measurement of unknown inductance in terms of known capacitor. Also derive the equation for quality factor.	**07**
	b.	Derive the equations for R^{th} and E^{th} for a slightly unbalanced wheat stone bridge.	**07**
7.	**a.**	Write short note on LVDT.	**07**
	b.	List and explain the factors that should be considered while selecting a transducer.	**07**
8.	**a.**	What is a data acquisition system? Mention the objectives of Data acquisition system.	**07**
	b.	With a neat block diagram, explain a generalized digital data acquisition system.	**07**